

Extension 1: Bank Loan System (IDs ending with 3, 6, 9)

You must add a new menu option: **9. Take a Loan.**

- Users can borrow any integer amount up to \$5,000.
- A user cannot take a new loan while another loan is active.

Loan Plans

After entering a **loanAmount**, the user must choose a plan. All calculations must use integer math and round up (ceiling) to the nearest integer.

1. Short-Term (3 Turns):

- Total Interest: 6% (Total to repay: (int) (**loanAmount** * 1.06))
- Payment per Turn: Total to Repay / 3 (rounded up)

2. Medium-Term (6 Turns):

- Total Interest: 12% (Total to repay: (int) (**loanAmount** * 1.12))
- Payment per Turn: Total to Repay / 6 (rounded up)

3. Long-Term (12 Turns):

- Total Interest: 24% (Total to repay: (int) (**loanAmount** * 1.24))
- Payment per Turn: Total to Repay / 12 (rounded up)

Repayment

You must modify your **endTurn** method. The loan payment is automatic and occurs *after* salaries are paid.

1. Add vehicle profits.
2. Subtract total salaries.
3. Subtract the **loanInstallmentPerTurn** (if active).

Your **endTurn** summary must now also print how much was paid to the bank, for example:

Loan Installment Paid: 517 \$

If **totalMoney** is negative after all payments (salaries and loan), the user is bankrupt and the simulation ends.

Extension 2: Staff Experience System (IDs ending with 1, 4, 7)

To reflect the growth of long-serving employees, you will implement an experience system. Staff who successfully complete their tasks will become more effective by improving their skills over time.

The Core Rules

1. **Gaining Experience:** At the end of each turn, staff members gain **1 experience point (XP)** based on their job:
 - **Receivers:** Gain 1 XP automatically every turn.
 - **Drivers & Carriers:** Only gain 1 XP if they were assigned to a vehicle that *successfully ran* (i.e., it had a driver and completed a delivery task).
2. **Salary Must Be Fixed:** This is a critical rule. The salary of a staff member **must not change** as they gain experience. Salary must be calculated *only* from the staff member's **original stats** at the moment they are hired.

How Experience Affects Skills

You must find a way to track the experience for each staff member and a way to store their original (base) stats.

When the simulation uses a staff member's skill, the new value should be their **(Original Base Skill) + (Bonus from Experience Points)**.

The bonuses are as follows:

- **Driver:** For every 1 XP gained, the Driver gets:
 - +1 to their `drivingSpeedLimit`
 - +1 to their `deliveryCapacity`
- **Receiver:** For every 1 XP gained, the Receiver gets:
 - +1 to their `receivingCapacity`
- **Carrier:** For every 1 XP gained, the Carrier gets:
 - +1 to their `priceBoost`
 - (The `deliveryCapacity` of a Carrier does *not* increase with experience.)

Extension 3: Contractor Staff System (IDs ending with 0, 2, 5, 8)

To provide more complex hiring strategies, you will introduce temporary "Contractor" staff, who have an upfront cost instead of an end-of-turn salary.

1. Modifying the Hiring Pool (in startNewTurn)

Your startNewTurn method must be updated. The hiring pool is now generated in two parts:

1. **Permanent Staff:** The system will *always* generate the **5 permanent staff members** as described in the original assignment.
2. **Contractor Staff:** In **addition** to the 5 permanent staff, the system will generate a *second* group of Contractors. The number of Contractors to be hired is **random (between 0 and 5)**.

Your total hiring pool each turn will now consist of 5 to 10 candidates.

2. Hiring Rules (in hireStaff)

You must clearly mark the employment type ([PERMANENT] or [CONTRACTOR]) when listing staff.

Hiring Limit: The original rule still applies, but it is now **more important**: You can hire a **maximum of 5 total staff members per turn**. This limit applies to the *combined pool* of 5-10 Permanent and Contractor candidates.

Hiring CONTRACTOR Staff:

- **Hiring Fee:** A one-time fee is paid *immediately* upon hiring. This fee is **60%** (rounded up) of what their normal salary would have been (`calculated_salary * 0.6`).
- **Salary (End of Turn): \$0.** Contractors do not receive a salary at the end of the turn.

3. End-of-Turn Rules (in endTurn)

Contractor staff are temporary and **only last for one turn**.

- At the end of the turn, *after* permanent staff salaries are paid, you must loop through your hiredStaff list.
- Any staff member identified as a CONTRACTOR **must be automatically fired and removed** from the hiredStaff list.